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Let the wind through: what Tacoma Narrows taught bridge design

The free guide to the reel on the 1940 Tacoma Narrows Bridge: why a moderate wind destroyed a thin solid deck, how engineers fixed it, and how to test a deck model with a fan.



What happened

The first Tacoma Narrows Bridge in Washington State opened on 1 July 1940. Its main span rested on solid plate girders only 2.4 m deep, which made it very slender and economical, and it moved in the wind from the start.

On 7 November 1940, in a wind of about 67 km/h, the deck began to twist from side to side, each swing larger than the last, until it broke and fell into the water. The camera of Barney Elliott recorded it; the film is in the public domain.

Why a moderate wind was enough

A solid, shallow deck behaves a little like a wing. Air cannot pass through it, so it flows around the edges and forms swirls (vortices) behind it. These push the deck up and down and make it rotate slightly.

When the deck twists, the angle of the wind on it changes, which pushes it even more in the same direction. Energy goes in faster than the structure can lose it, so the twisting grows. Engineers call this aeroelastic instability, or flutter. The wind does not need to be strong, only steady.

The fix

The replacement bridge, opened on 14 October 1950, uses a deep open stiffening truss and openings in the deck so that air can flow through instead of around it. Since then, long bridge decks are tested as scale models in wind tunnels before they are built, and many use streamlined box shapes or slots that keep the air flow calm.

Built examples

Tacoma Narrows Bridge (1950)	Deep open truss and deck openings; a parallel bridge was added in 2007.
Golden Gate Bridge, San Francisco	After strong winds made its deck move in the early 1950s, lateral bracing was added under the deck to stiffen it against twisting.
Akashi Kaikyo Bridge, Japan (1998)	Before construction, a large scale model of the whole bridge was tested in a wind tunnel; its truss deck includes devices that steady the air flow.

Try it yourself: a deck in the wind

1. Cut two strips of thin card, 40 cm long and 6 cm wide. Leave one solid; punch a row of holes or cut slots along the middle of the other.
2. Hang each strip between two piles of books with thread at both ends, so it can twist a little, like a bridge deck on its cables.
3. Place a desk fan 50 cm away, level with the strip, and start at the lowest speed.
4. Watch the solid strip: at some speed it starts to rock and twist, more and more. Note the fan setting.
5. Repeat with the slotted strip at the same settings: it moves much less, because air passes through.
6. Try a third strip folded into a shallow triangle (a streamlined shape) and compare all three on video.

The rule to remember

Slender and solid can look elegant and still fail: let the wind through, and test the shape before you trust it.

Default answer	Make it thinner and smoother because it looks better on the drawing.
Form-finder answer	Open the section, streamline it, and test a model in moving air.
Check	Ask: what does a steady, moderate wind do to this shape over ten minutes?

Go further

Tacoma Narrows Bridge (1940), Wikipedia
https://en.wikipedia.org/wiki/Tacoma_Narrows_Bridge_(1940)

HistoryLink: Tacoma Narrows Bridge collapses
https://historylink.org/File/5048

Britannica: Tacoma Narrows and bridge engineering
https://britannica.com/technology/bridge-engineering/Tacoma-Narrows

Credits

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Facts: HistoryLink, Britannica, Wikipedia.

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